

# Grade 4 lessons

## Big multiplication, factors and decimals

### IN THIS BOOK

1. Bigger multiplication
2. Division with remainders
3. Factors, multiples and primes
4. Equivalent fractions
5. Decimals

### HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 4 Kahoot and Blooket sets, and one stop on the Grade 4 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

# Bigger multiplication

## THE BIG IDEA

Break a big multiplication into smaller parts that you already know, then add the parts together.

## HOW TO DO IT

1. Split one number into tens and ones.
2. Multiply each part by the other number.
3. Add the answers.
4. To multiply by 10, 100 or 1000, write that many zeros on the end.

## WORKED EXAMPLES

**$34 \times 6 = ?$**

$30 \times 6 = 180$  and  $4 \times 6 = 24$ . Add them:  $180 + 24 = 204$ .

**$23 \times 14 = ?$**

$23 \times 10 = 230$  and  $23 \times 4 = 92$ . Add them:  $230 + 92 = 322$ .

## WATCH OUT

Remember the zero when you multiply tens.  $30 \times 6$  is 180, not 18.

## WORDS TO KNOW

**partial products**

the smaller answers you add together

# Division with remainders

## THE BIG IDEA

Sometimes a number does not share out evenly. What is left over is called the remainder.

## HOW TO DO IT

1. Ask how many times the divider fits into the number without going over.
2. Multiply to see how much that uses.
3. Subtract to find what is left. That is the remainder.
4. Write the answer with R, like 3 R2.
5. Check: divider  $\times$  answer + remainder gives the starting number.

## WORKED EXAMPLES

**$17 \div 5 = ?$**

5 fits in 3 times:  $5 \times 3 = 15$ . Left over:  
 $17 - 15 = 2$ . The answer is 3 R2.  
Check:  $5 \times 3 + 2 = 17$ .

**$47 \div 6 = ?$**

$6 \times 7 = 42$ . Left over:  $47 - 42 = 5$ . The  
answer is 7 R5.

## WATCH OUT

The remainder must be smaller than the number you divide by. If it is not, the divider fits in one more time.

## WORDS TO KNOW

**remainder**

what is left over after dividing

**divisor**

the number you divide by

# Factors, multiples and primes

## THE BIG IDEA

Factors are numbers that divide evenly into a number. Multiples are what you get when you skip count. A prime has only two factors: 1 and itself.

## HOW TO DO IT

1. To find factors, look for pairs that multiply to the number. Start with 1 and work up.
2. To find multiples, skip count: 3, 6, 9, 12 and so on.
3. To test for prime, try dividing by 2, 3, 5 and 7. If none divide evenly (for numbers under 100), it is prime.
4. The least common multiple is the first number that appears in both lists of multiples.

## WORKED EXAMPLES

### Find all the factors of 12.

Pairs:  $1 \times 12$ ,  $2 \times 6$ ,  $3 \times 4$ . The factors are 1, 2, 3, 4, 6 and 12.

### What is the least common multiple of 4 and 6?

Multiples of 4: 4, 8, 12. Multiples of 6: 6, 12. The first one they share is 12.

## WATCH OUT

1 is not a prime number. 2 is the only even prime.

## WORDS TO KNOW

**factor**

a number that divides evenly into another

**multiple**

a number in the times table of another

**prime**

a number with exactly two factors

# Equivalent fractions

## THE BIG IDEA

Different fractions can name the same amount.  $\frac{1}{2}$ ,  $\frac{2}{4}$  and  $\frac{3}{6}$  are all equal. They are called equivalent fractions.



$\frac{2}{4}$  covers the same amount as  $\frac{1}{2}$ .

## HOW TO DO IT

1. Multiply the top and bottom by the same number to get an equal fraction.
2. Or divide the top and bottom by the same number to simplify.
3. To compare two fractions, give them the same bottom number first.
4. Then compare the top numbers.

## WORKED EXAMPLES

$$\frac{2}{3} = \frac{\square}{12}$$

The bottom was multiplied by 4 ( $3 \times 4 = 12$ ). Multiply the top by 4 too:  $2 \times 4 = 8$ . So  $\frac{2}{3} = \frac{8}{12}$ .

**Which is bigger,  $\frac{3}{4}$  or  $\frac{2}{3}$ ?**

Use twelfths.  $\frac{3}{4} = \frac{9}{12}$  and  $\frac{2}{3} = \frac{8}{12}$ . 9 is more than 8, so  $\frac{3}{4}$  is bigger.

## WATCH OUT

Do the same thing to the top and the bottom, and only multiply or divide. Adding the same number does not work.

## WORDS TO KNOW

**equivalent**

equal in value

**simplify**

write with the smallest numbers possible

# Decimals

## THE BIG IDEA

Decimals are another way to write fractions with 10 or 100 on the bottom. The first place after the point is tenths. The second is hundredths.

## HOW TO DO IT

1. Read 0.7 as 7 tenths. That is  $\frac{7}{10}$ .
2. Read 0.25 as 25 hundredths. That is  $\frac{25}{100}$ .
3. To compare, line up the decimal points. Add a zero if it helps: 0.5 is the same as 0.50.
4. To add or subtract, line up the decimal points and work as usual.

## WORKED EXAMPLES

### Which is bigger, 0.5 or 0.45?

Write 0.5 as 0.50. 50 hundredths is more than 45 hundredths. So 0.5 is bigger.

### 2.3 + 1.8 = ?

Tenths:  $3 + 8 = 11$  tenths. Write 1 and carry 1. Ones:  $2 + 1 + 1 = 4$ . The answer is 4.1.

## WATCH OUT

More digits does not mean a bigger number. 0.45 is smaller than 0.5.

## WORDS TO KNOW

**tenth**

one of ten equal parts

**hundredth**

one of a hundred equal parts